THE THESIS BEHIND THE ENERGY TO COMPUTE FLOW TERMINAL

The Electrons Must Flow

What I learned mapping companies from the turbine to the token.

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> ***"The spice must flow."***
>
> DUNE

Frank Herbert's Dune is remembered for its deserts, its religion, and its politics. Underneath all three it is a book about an economy that runs on a single flow. Herbert imagined a universe in which travel, trade, and power all depended on one substance from one planet, then followed what that dependence does to prices and politics. His answer fits in four words, repeated like a law of nature. The spice must flow. *(The exact words are from the 1984 film rather than the novel, but no line sums up what the book is about better.)* Everyone says it because everyone's power depends on it. And the logic underneath those four words transfers exactly to the energy-to-compute market. When everything depends on a flow, power belongs to whoever stands where it narrows.

The flow I watch runs from electricity to intelligence. The clearest way to see what a position on it is worth is to look at what people pay for time on it.

So let's look at two contracts, signed a few months apart. Anthropic signed a twenty-year lease worth roughly nineteen billion dollars in contracted revenue for a Kentucky campus that does not exist yet. That is a patient buyer, planning ahead. Reflection agreed to pay SpaceX $150 million a month for immediate access to ready GPUs at Colossus 2 in Memphis. That is an impatient buyer, paying now. Work both out per unit of energy, as the power analyst Neel Somani brilliantly did, and the patient buyer pays about $271 per megawatt-hour while the impatient one pays the equivalent of nearly $5,000.

The comparison is not clean, and Somani says so himself. One price includes racks of ready GPUs while the other covers only the building and the cooling around them, which is why he calls his own number napkin math. Subtract the hardware, though, and a large premium survives. And once it survives, ask yourself what the two contracts are actually pricing, because both of them buy the same electrons. Not electrons. When. In this market the hardware is written off with barely a shrug. Reactors get abandoned mid-permit and GPU fleets depreciate in three years, and through all of it nobody writes off time. Every dollar in this essay is, one way or another, a price paid for time.

I wanted to understand who gets paid in a market that prices time like that. The problem is that the question runs through half a dozen fields at once, siting and permitting, grid engineering, nuclear licensing, power electronics, thermodynamics, chip packaging, market structure. Nobody is an expert in all of them and I am certainly not. So I did the thing I could do. I broke the flow from electricity to intelligence into eleven stages, from siting and permits at one end to compute orchestration at the other, and mapped the early-stage companies forming at each, more than 450 of them. It is a map of the private layer, where founders are still choosing stages, and it skews North American. It is not complete and it does not need to be. The point was never the list. The point was to sort the buildout by physics instead of by market category and then count, and a base this size is enough for the patterns to show.

What is the buildout, physically?

Strip away the logos and it is a single process. Somewhere in west Texas, at this moment, an electron is leaving a turbine. It departs at 18,000 volts and is pushed up to 345,000, because high voltage is how electricity travels far without losses. It crosses a hundred miles of transmission line, drops through a substation, and threads a transformer the size of a school bus. It enters a building that did not exist two years ago. Inside, it falls to 800 volts in the newest halls, then to around 50, then below one, descending conversion after conversion like a river descending locks, and each lock spills something. It arrives at a processor no wider than a playing card and exists there, as useful energy, for a few nanoseconds. In that instant it flips transistors that multiply enormous matrices of numbers, and those multiplications, stacked layer on layer, become a prediction of the next word of someone's question. Then the electron is gone. A token leaves the building as light down a fiber, and somewhere far away, a sentence grows one word longer.

(A physicist will object that no single electron makes this trip, and the physicist is right. Electrons in a wire drift at a walking pace measured in millimeters per second, and what actually travels, at a good fraction of the speed of light, is the energy in the field around the wire. That said, the fiction is worth the clarity, so I will keep it.)

That is the trip in volts. Now take the same trip again in the unit this essay actually cares about, which is how long each thing the electron passed through took to exist.

The parcel the building sits on was found by consultants working site by site, a few months per site, and its zoning and permits ran six to eighteen months depending on the county. The request to connect it to the grid went into a utility queue. For power plants, the median project built in 2025 had waited 61 months from its interconnection request to commercial operation, about 59 for anything above 200 megawatts, up from 36 months in 2015 and 22 in 2008. Large loads are newer to queues and in the busiest markets are already slower. Dominion Energy Virginia has about 70 gigawatts of data-center requests waiting against an all-time system peak of 24.7 gigawatts. It has given 25 of those gigawatts connection dates running out to 2031 and the other 45 no date at all. The transformer at the substation was ordered about 128 weeks before it arrived, by Wood Mackenzie's mid-2025 survey. The step-up transformer at the plant was worse and getting worse, past 160 weeks by the first quarter of 2026 against an average of 143 weeks in 2024. The high-voltage circuit breakers beside them ran 125 weeks in the second half of 2025, up from 77 in 2023. If the site generates its own power, the three companies that make heavy-duty gas turbines are booking slots four to five years out, and GE Vernova is taking reservations for 2031 delivery with orders and reservations totalling 116 gigawatts as of the second quarter of 2026. If it wanted nuclear, the first commercial advanced reactor to clear the regulator, TerraPower's Natrium plant at Kemmerer, Wyoming, spent more than four years working with the regulator in all, filed its application in March 2024, and had its technical review finished in eighteen months, nine months ahead of the regulator's own schedule, before the construction permit arrived in March 2026 for a plant whose construction is due to finish in February 2031.

The building itself, the part everyone pictures when they say construction, took eighteen to thirty months from groundbreaking to energization, and the average campus phase now runs 22 months, fifteen percent longer than in 2023, with almost all of the stretch coming from electrical equipment rather than concrete. The accelerator inside waited on a packaging slot at TSMC with a lead time of 52 to 78 weeks, and on high-bandwidth memory whose 2026 output was sold out before the year began. And when the chip finally arrived, the software that makes it usable was months of specialist work per architecture.

Read the ledger and one thing stands out. In a favourable market the development cycle alone, from first site visit to the day the money is committed and before a shovel moves, runs eighteen to thirty months. In a constrained one it runs four to seven years, and then the building still has to be built. The electron's own journey took a fraction of a second. The journey that made its journey possible took half a decade, and almost none of that half-decade was spent pouring concrete. It was spent waiting, in queues held by utilities, factories, and regulators. That is what the two contracts at the top of this essay were pricing. Anthropic bought a place at the front of a five-year line and paid $271. Reflection bought the line's absence and paid $5,000.

Two facts about this flow set the terms for everything else.

First, it is lossy, and it is worth a paragraph on why, because the reason comes back at three of the eight stops. Every conversion runs through a switch, and a switch is never perfectly on or perfectly off. While it is on it has resistance, so it heats. While it is changing state, for a few tens of nanoseconds, voltage and current overlap across it and that overlap is dissipated as heat too, and it happens on every cycle, so the loss rises with how fast the switch runs. Around the switch sit the inductors and transformers that smooth the output, and those lose energy in their windings and in their cores. A well-designed stage recovers 96 to 98 percent of what enters it, which sounds like nothing until you chain five of them and the compounding takes you below 90. That is where NVIDIA's own 800-volt materials put today's end-to-end chain, with industry estimates running from the low eighties up. Then the heat has to be hauled back out. The industry's average power usage effectiveness is 1.52, which means for every watt reaching a server another half watt goes to cooling and overhead, and that figure has drifted down only slowly for a decade. Weight it by capacity and it improves to 1.36, and that number has been improving faster since 2024 as the biggest new halls come online. The best hyperscale sites run near 1.1. Put conversion and cooling together and, at a good site, roughly three quarters to four fifths of the power at the fence reaches the IT equipment, which then converts it again on the board before any of it becomes a token.

Second, the flow is only as fast as its narrowest point. It does not matter how many GPUs you own if your grid connection is in year three of study, and it similarly does not matter how fast your chip is if it is waiting on memory. The flow has one throughput, set by whichever stage is currently the bottleneck. And the bottleneck moves like the worm beneath Herbert's sand. You feel where it surfaces, and you listen for where it will surface next. Three years ago it was chips. Then transformers. Then interconnection. Then cooling. Then memory. Right now, in different places, it is all of them at once.

That is the physics. Here is the frame it forced, which I ended up testing everything against. In a lossy flow with moving bottlenecks, the interesting opportunities are the ones that either shorten the flow or own a point it cannot route around.

What are the two moves, exactly?

The two moves deserve careful definitions, because the rest of the essay leans on them. And the first thing to notice is that they are different kinds of claim. Shortening is a bet about physics. Owning is a bet about power. The whole game, as we will see, is turning the first into the second.

A company shortens the flow if, holding everything else constant, its existence means more tokens come out per joule, per dollar, or per month of waiting. The test is physical. Delete the company from the world and ask whether the flow gets longer, slower, or leakier. Shortening comes in three forms.

Remove losses. Recover part of the energy that never reaches the chip.

Remove steps. Delete a conversion, a study, a manual process entirely. This is where a question that ran under my whole walk lives, which parts of the accepted path are physics and which are history. Solar panels produce direct current. Batteries store direct current. Chips consume direct current. Everything between those endpoints, the inversions and rectifications and step-ups and step-downs, is the standard path, and it was built for a different problem, moving alternating current long distances to spinning motors. History can be deleted. Physics cannot. NVIDIA's 800-volt roadmap is an exercise in deleting history. Today's data hall hands power through a medium-voltage transformer, then a UPS that turns AC into DC and back into AC again, then the distribution gear, before the rack's own power supplies finally make the DC the chips wanted all along. The roadmap replaces all of that with one conversion, grid AC to 800 volts DC at the building's edge. Inside the rack it goes further. Where today's tree steps 400 volts down to 50 and then 50 down to 12 in two separate converters, a single 64-to-1 converter takes 800 straight to 12, and NVIDIA's white paper prices the deleted stage at an extra point of efficiency and 26 percent less board area in the crowded space beside the GPU. These are design targets for racks due in 2027 rather than shipped hardware, and I treat them as a calendar rather than a result.

Herbert ran the same test on his own universe. Arrakis held its price for exactly as long as nobody could make spice anywhere else, and the moment the Tleilaxu learned to grow it in a tank, later in the series, the planet's monopoly was finished. A stage on the flow works the same way. If it exists because nothing can replace it, it holds its price. If it exists because nobody has replaced it yet, someone will, and the AC conversion chain is the clearest case on the map of a stage in the second category.

Remove waiting. Compress a queue, a bring-up, a permit.

All three count, and the market prices them very differently. Recover a loss and you save percentage points of energy cost. Delete a step and you save its loss plus its equipment queue. Delete waiting and you move the project's entire revenue forward in time. Right now that is the largest effect of the three, and the arithmetic says by how much. Take a one-gigawatt campus. A point of conversion loss recovered is about 88,000 megawatt-hours a year, which at fifty dollars a megawatt-hour is roughly four million dollars. A year of waiting removed is a year of the campus's revenue pulled forward, and at the $271 a megawatt-hour Anthropic agreed to pay for a building and its cooling, a year of a gigawatt is about $2.4 billion. That is the spread at the top of this essay, the same megawatt-hour selling for $271 to a patient buyer and nearly $5,000 to an impatient one, and it is the market saying that today the queue is what binds.

But the ranking is a statement about this moment, and two things complicate it. It flips over time, for one. Waiting is worth the most while the queues are five years long, and interconnection reform, new transformer plants, and turbine capacity are all working to shorten them. The four million a year is paid every hour the campus runs for twenty years and does not care what the queue is doing. When the queues clear, the loss and the step become the durable value. And the size of an effect and who captures it are different questions, which for a startup is the one that matters more. The $2.4 billion of pulled-forward revenue lands with the developer who owns the campus, and the company that removed the wait sells into a market where the next shortener can do the same. A deleted conversion stage is different. It is a component that ships in every rack for the life of a product generation, it is designed in once and then very hard to design out, and that is how a shortening of the second kind turns into a position while a shortening of the third kind usually does not. So read the three forms as a question of when as much as which. Waiting pays most today and mostly to someone else. Steps and losses pay less per year, pay for longer, and are the forms a component maker can actually own.

And be clear about what fails the test. A marketplace renting you the same GPU more cheaply has moved money between pockets while the flow stayed exactly as long as it was. Software that produces a prettier permit application with no earlier approval date has shortened nothing. A consultancy that helps you wait more knowledgeably is selling commentary on the queue.

A company owns a chokepoint if the flow cannot route around it, and the value of the position follows one rule. A chokepoint charges what the detour costs. If every alternative path costs a competitor two years, the tollbooth on the short path can charge up to two years of value, and the traffic pays. At a chokepoint, the position is the moat.

Plenty of things look like chokepoints without being one, and the difference is what the position is made of. Real chokepoints are made of physics. High-bandwidth memory is one, because only three companies on earth can fabricate it, and what keeps a fourth out is the stacking and packaging yield that every new entrant would have to learn from zero. Enriched fuel is another, because it requires a licensed facility. Energized land is a third, because its grid connection took five years to win. Or they are made of law, an exclusive license, an approved design, a seat in a regulated market. Herbert knew this one too. All Imperial commerce routed through one company, CHOAM, and the real prize in his universe was a directorship. They can also be made of institutional process or specialized talent, a fab's yield learning, a utility's vendor qualification, knowledge that lives in feedback loops rather than in any one person's head. So when someone claims a moat, ask three questions. What is it made of? What is its half-life? And what is it, in turn, hostage to? A transformer queue is a chokepoint today and is being competed away as we speak. Enriched fuel and HBM are structurally harder to route around. And even the hardest positions hang from something upstream. HBM is hostage to advanced-packaging capacity and to new DRAM fabs, and a new DRAM fab is hostage to construction crews and a grid connection, which is where this essay began. The detour always gets cheaper eventually. The question is whether it takes two years or twenty.

One more thing, and it took me the longest to see. The two moves are less alternatives than a sequence, and the sequence is how a physics bet becomes a power position. Shortening creates value but struggles to keep it, because a shortener sells into a competitive market for the thing it shortens. The time it saves is mostly captured downstream, by whoever owns the asset that starts earning sooner. Owning captures value but rarely offers an opening, because the good chokepoints are mostly claimed. So the companies I find most interesting run the sequence. They shorten to enter, and they let the shortening accumulate into something ownable, a dataset, a delivery slot, a qualified-vendor status, an installed control plane. Ask every shortener what the saved time compounds into. If the answer is a stronger sales pitch, it is a supplier having a good quarter. If the answer is a position, it is a chokepoint in formation.

Where are the founders actually showing up?

One thing the map can show is where new companies are actually forming along the flow, and I expected the answer to track where the bottlenecks are. It doesn't. Formation clusters at the two ends. In siting and permits, 53 percent of the stage's 41 companies were founded in 2024 or later. In compute orchestration, 54 percent of its 81. Through the middle of the flow, transmission, transformers, switchgear, construction, conversion, the share runs between the teens and low thirties. The two end stages hold 27 percent of the map's companies and took 46 percent of everything founded since 2024, about four new companies at each end for every one in each middle stage.

Part of that is no surprise, and I want to say so before drawing anything from it. The middle is where the contracts are large, the buyers are utilities, and the products are regulated hardware. That is incumbent territory by construction. The companies serving it are old, often public, and were never going to appear on a map of startups. The ends are software-shaped. A siting company is data and a workflow. An orchestration company is a control plane. Small teams can form there in a quarter. So the dumbbell measures where a startup can be founded at least as much as where the pain is, and a reader who takes it as a map of the bottlenecks would be reading it wrong. The map is also incomplete. It is a private layer I assembled rather than a census, it skews North American, and I am certain there are more companies in every stage than I have found, especially in the middle, where a 2019 test lab in Ohio is harder to find than a 2024 launch post. That work continues and the terminal will keep growing.

What the shape is useful for is the mismatch it exposes. The bottlenecks this essay has been describing sit in the middle. The transformer queue, the interconnection study, the switchgear lead time are all in stages where almost nobody is founding companies. The founders are at the two ends, and the ends are crowded. When half of a stage's companies are under two years old, that stage's window is already closing and its winners will be decided in the next eighteen months. So the dumbbell says two different things to someone writing seed checks. At the ends, the race is on, and the question is who compounds a shortening into a position first. In the middle, nobody builds a transformer factory on a seed round, so the way in is to shorten around the incumbent or to supply it. Ayr routes transformer orders to Indian factories that never stopped growing and hands its customers two years. Voxel deletes the AC conversion stages a self-powered site never needed. Vertical Semiconductor makes the switch the power-supply incumbents' own roadmap requires. Those are the three shapes a startup takes in the incumbent middle, and the walk will show each of them.

The other thing the map is good for is what it leaves blank. Combined with the timing patterns in the next section, the empty columns start generating predictions. Some are already coming true as stops on the walk, answered by companies this essay features. Others are still empty. Water as procurement rather than cooling, the rights, reuse, and acquisition layer that already decides permits in Texas, has no seats taken at all. Nuclear's operational layer, the licensing throughput, fleet software, and qualified supply chain a family of small reactors will need, has exactly one early arrival. And those are just a few of them. The map holds more.

When does a stage open, and when does it close?

Two more patterns from the map, and both are about timing, which is why they belong together. The first says when a stage stops being investable. The second says when it starts.

Stage by stage the same life cycle repeats, and it is worth slowing down on the mechanics because this pattern does more work in the walk than any other. A bottleneck bites somewhere on the flow. A cohort of startups forms to attack it. Capital floods in. Then the cohort resolves into its winners, and how many there are depends on the stage. Where the product is a control plane, a standard, or a reference price, the market usually tolerates one, because the value of the position is that everyone routes through it. Where the product is a component, buyers dual-source on principle and the stage settles on two or three, which is what happened in cooling, where CoolIT went to Ecolab, Boyd to Eaton, and Motivair to Schneider inside eighteen months. Either way the signal is the same. The winners get acquired at a scale no seed check can follow, or they raise at valuations that already price the problem as solved. That is what I mean by a window closing. The pricing event closes it, however many survivors there are. The chance to back a small company against that specific problem is gone, because the problem is no longer open, and backing one anyway means paying billions for certainty that was available at seed prices three years earlier.

But here is the part that makes closed windows useful rather than just sad. The winners actually fix the stage, and once a stage is fixed it stops being the thing that limits the flow. The limit reappears at the next weakest stage beside it, where there are no winners yet and the companies are still small.

Cooling ran the full cycle in under four years. GPUs got too hot for air, a cohort formed, direct-to-chip liquid cooling won, and Ecolab's $4.75 billion for CoolIT in March 2026 was the settlement. That closed the chip-level window, and it moved the problem rather than ending it. Liquid loops unlocked denser racks, the heat still has to leave the site, and the constraint now sits at the facility boundary, where a new cohort is forming at seed prices.

The map keeps more receipts like that, and each one reads the same way. What is closed, and where does it point? RadixArk, the company formed by the creators of the SGLang inference engine, launched in May 2026 with a $100 million seed at a $400 million valuation. SGLang's central trick is reusing the memory cache across requests, so that the same GPU memory serves more tokens, and a seed round at that price says the software route to wringing more out of existing memory is already priced as solved. The distinction matters, because software that wrings more out of existing arithmetic, the kernel and compiler work that makes a new chip usable or a deployed one run at full speed, is a different problem and is still wide open, as the last stop on the walk will show. What RadixArk's price closes is the memory door specifically. That points down, at the physical doors through the memory wall, the wire, the package and the stack, where three young companies on the map are attacking the same wall from three different layers.

General Matter holds one of three $900 million Department of Energy contracts, awarded in January 2026, to build enrichment capacity for advanced-reactor fuel at Paducah, Kentucky. And Aalo brought a full-scale test reactor from groundbreaking to first criticality in under eight months at Idaho National Laboratory on July 4, 2026. The fuel chokepoint is being claimed and the reactor cohort is keeping its calendar, which together point at the layer neither of them is building, the licensing, siting, and operations software a fleet of small reactors will run on. And Compute Desk and Nodal Exchange announced in September 2026 that they will list CFTC-regulated GPU compute futures settling against Compute Desk's indexes, with CME and ICE preparing rivals. The reference-price seat in compute is being contested right now, and the contest points beneath it, at the physical layer, because paper only trades as well as the underlying compute can actually be delivered and moved. In every case the interesting question is not who won the closed stage. It is where the constraint went, because that is where the next cohort is forming while it is still cheap.

The second timing pattern runs the other direction. The industry announces its own future in writing, years ahead, and then finances the companies that must exist to meet it. When a dominant player publishes a new spec, it simultaneously publishes a list of components that do not exist yet, and a dated cohort of companies forms within quarters to build them. NVIDIA's 800-volt architecture is the clearest case. The published roadmap named the power-electronics parts the transition requires, wide-bandgap switches, solid-state transformers, 800-volt-to-sub-one-volt converters, and a cohort formed to build exactly those. The co-packaged-optics roadmap is doing the same thing now for photonic interconnect. And the spec-setter does not just publish the calendar, it finances it. By my count, NVIDIA sits on 18 cap tables across eight of my eleven stages, with dozens more companies carrying its imprint through Inception and reference designs. When the spec-setter tells you what must exist next, believe it.

Two updates keep this honest. The calendar is no longer one company's. The 800-volt work is now published jointly with Google and Microsoft through the Open Compute Project, with an OCP white paper on low-voltage DC distribution dated March 2026 from a working group including Google, ABB and Siemens, a draft solid-state transformer design specification written by the three companies and published in July, and more than eighty companies building to it, which makes the dates more binding than any one vendor's roadmap could. And there is a bear case worth stating plainly. If the spec-setter publishes the calendar, finances the cohort, and controls the reference design, some of these companies are less startups than a supplier-development program wearing venture pricing, with exits capped at strategic-acquisition prices. The calendar de-risks timing. It does not promise the value lands with the cohort.

So the frame has three parts, and they nest. The two moves say what a company has to do. The dumbbell says where on the flow to look for it. The two timing patterns say when a stage is open. Now let's take them and walk. Eight stops, each framed the way I actually encountered it, as a question.

Stop one: can software release megawatts that already exist?

A data center developer who needs a gigawatt for a campus runs into three problems, in a fixed order, and at each of them the power already exists while the developer waits.

The first problem is where to build. The developer needs to know which parcels the grid can serve, and that information exists today. Every utility knows how loaded each of its substations and feeders is, the county knows the zoning, and fiber routes and water rights are on maps. Nobody has put those records together in one place, so the developer hires consultants to assemble them parcel by parcel. A desktop diligence report on a single site costs about $25,000 and takes six weeks, and a campus search screens eight to fifteen sites before it settles on one. That screening alone takes months, and it happens before a single request has been filed with a utility. Nothing in it requires new information, only the assembly of information that is already there.

The second problem is the utility's answer. The developer files a request for a hundred megawatts or more, and the utility has to work out what that load will do to its grid. That used to be quick, because loads were motors and heaters and a steady-state model of the power flow told the engineers everything they needed. A data center is different. It can drop offline in a single cycle, and a training cluster swings tens of megawatts every few seconds, so the utility now has to run dynamic stability studies as well, and those studies need data about how the load behaves that the developer cannot supply this early. So the engineers assume the worst case, and the worst case means new lines and substations, and new lines mean years. The studies also queue. Dominion Energy Virginia works through its requests ten at a time in the order they arrived, and with seventy gigawatts waiting against a system peak of twenty-five, the data center industry's own trade group told Virginia's regulator that a project can wait fifteen years. The problem belongs to the developer, but the process belongs to the utility, so whoever shortens this step has to do it from inside the utility.

The third problem is the power itself, and it comes in two forms. The first form is the date the utility finally gives, and it is often 2031 or later. The utility studied the campus as a load that draws its full gigawatt in every hour of the year, including the one hour when the grid is most stretched, because that is how utilities have always planned. So the grid is built for that hour, and in an average hour nearly half of it is idle. Duke's Nicholas Institute asked how much new load could fit on the existing grid if that load agreed to turn down during the stretched hours. The answer in Rethinking Load Growth was 76 gigawatts across the 22 largest grid regions if the load is curtailed a quarter of one percent of the time, and 126 gigawatts at one percent. The average event lasts about two hours, and in most of them the load keeps at least half its power. For scale, 126 gigawatts is more than a hundred large power plants. The study leaves out transmission limits, so some of that headroom sits where no line can carry it to a campus, and the authors call it a bridge, because once demand grows into the idle hours the room is used up. Neither changes the reason the date was wrong. The date assumed a load that never bends, and a data center can turn down in seconds.

The second form is the power the developer brings. Because the date is 2031, most campuses breaking ground now plan their own generation behind the meter, gas turbines, engines and fuel cells for the first years with batteries beside them, and the grid connection to follow. That solves the date and creates a new job, because somebody now has to run a small power plant, a battery, a grid connection and a load that swings tens of megawatts every few seconds, all at once. When the utility calls for a turn-down, the jobs on the GPUs have to throttle. When a turbine trips, the batteries have to carry the load for the seconds it takes to shed it. When the communication phase of a training step drops the draw, the batteries should charge, so that the turbines can be sized for the average rather than the peak. Each of those is a control decision taken in seconds across assets that today have separate vendors and separate screens, and without it the developer buys turbines for the peak of the swing and lets the batteries sit idle between steps. The megawatts here are the developer's own, already paid for and unused. A campus that can make those decisions is a flexible load in the utility's eyes and a cheaper plant in its own, and nobody sells that control layer off the shelf yet.

Until this year the flexible half of that was an argument, and this year it became a market. In March EPRI's DCFlex program published Flex MOSAIC, five standard classes of load flexibility that a utility planner can put in a model, signed by NERC, MISO, SPP, Google, Meta and NVIDIA. Google has written a gigawatt of that flexibility into its utility contracts, and Indiana's regulator approved the first one because the promise to turn down removed a grid upgrade other ratepayers would have paid for. In June FERC gave all six grid operators sixty days to justify or rewrite their large-load rules, and named a new service for flexible loads as one of five reforms it wants. A load that can prove it flexes now has somewhere to sell that proof.

So the gigawatt was there at every step, and all three problems are information and control inside a process the utility runs. Software is the only tool that reaches in there, and that is the shortening at this stop. The chokepoint is the utility's signature, because nothing energizes without it, and the company a utility comes to trust at any of these three steps holds a position the next entrant cannot buy.

Three companies have each taken one of the three problems, and it is worth seeing how each arrived at it.

Build took the first. Its co-founder James Stirrat-Ellis is an architect who worked at Heatherwick Studio on Changi Airport's Terminal 5 and on Coal Drops Yard in London, and somewhere in that work he came across a desktop diligence report for a data center site, a $25,000 PDF that had taken a team of specialists six weeks, and saw that every step in it could be automated. With Ben McClusky, whose background is multi-agent AI research, he built that. Build calls itself an agentic development firm, which in practice means thousands of AI subtasks running in parallel on site sourcing, power and grid-impact modelling, diligence and permitting, with every deliverable checked by a senior human reviewer before it goes out, and the customer paying for the finished decision. Underneath the agents sits the asset that makes them work, a parcel-level map of power capacity, substation loading, fiber and zoning across eighteen countries, updated continuously and cited in every output. It has delivered more than 150 projects for developers, operators and public bodies, including Tishman Speyer, which liked the work enough to become an investor as well as a customer. Take Build out of a project and the screening months come back, which is the test this essay applies to shortening. The map underneath is what the shortening compounds into, because every engagement deepens it and a later entrant would have to buy or rebuild it.

Senpilot took the second, from the only side that can fix it. The years lost at this step are lost inside the utility, in interconnection studies, capital plans, rate cases and regulatory filings, all produced by engineering teams staffed for a load-growth era they never planned for. Senpilot builds AI agents for those workflows, and its leadership is drawn from the people who spent their careers selling utilities their systems, a president who ran the utilities businesses at Oracle Utilities and GE Vernova, and advisors who held nearly every operating role at Toronto Hydro. Few companies stand where it stands, because the buyer is a utility and utilities are the slowest buyers in the economy. Its headcount has roughly tripled in a year. The interconnection study cannot be shortened from outside, so faster studies and faster filings are the only lever that reaches that waiting, and Senpilot is building the lever for the one buyer that can pull it. It becomes a position if utilities come to run their studies through it by default.

Soma Energy took the third, on live assets rather than paperwork. Its CEO, Ath Caramanolis, created the renewable energy optimisation team at AWS and scaled it to about ten gigawatts, so he spent years inside the largest corporate energy buyer on earth watching where flexibility went unused. His team now runs one AI platform on both sides of the meter. For power producers it decides when to generate, store or trade, with about two gigawatts under optimisation so far. For data centers it runs the behind-the-meter side as one control layer, the on-site generation, the batteries and the workloads themselves, so that jobs shift in time or location when the grid asks and the batteries buffer the swings in between, which is the job described above. Sitting on both sides matters, because the platform that sees how producers dispatch and how loads can flex is the one positioned to write a flexibility class into a contract and then honour it when the utility calls. Every megawatt of flexibility it can prove is a megawatt of the Duke headroom the grid can serve without building anything, which is the Duke arithmetic turned from a white paper into a P&L.; The shortening is deleted waiting at the largest scale on offer. The ownership is the installed base, because a platform that already dispatches two gigawatts of generation and runs the batteries and jobs on a campus is a system a later entrant would have to displace rather than copy.

Stop two: why can't a data center just buy a reactor?

Say the developer from stop one decides not to wait for the grid and wants firm power on its own land. Gas is the immediate bet and nuclear is what comes next. What the developer wants from nuclear is simple to state. A hundred-megawatt block, built in a factory, delivered in two years, sold as power by someone else who runs it. That is how a gas turbine is bought today. Nothing like it exists in nuclear, and the reason leads to one component.

Large reactors already serve data centers at both ends of the range. Existing plants are being contracted as fast as they can be found, Microsoft is restarting Three Mile Island under a twenty-year deal and Amazon is drawing up to 1.9 gigawatts from the plant next to its Pennsylvania campus, and the pool of such plants is small and nearly spoken for. New large plants are being planned for gigawatt campuses, and the earliest of them target the early 2030s, so the gigawatt-scale supply before then is restarts of plants that already exist. For a hundred megawatts to a gigawatt on the developer's own land inside this decade, the machine has to be small, and that is the machine this stop is about.

Start with the fuel and the regulator, because they narrow the field. The Energy Information Administration sorts every American small reactor design into four families. One family is light-water reactors. These are smaller versions of the plants running today, cooled by water and fuelled with ordinary low-enriched uranium. The other three families are cooled by gas, molten salt or sodium, and most of them run on HALEU, uranium enriched to just under twenty percent rather than the five percent in ordinary fuel. Those choices may well be the right ones for the long run. They are slow ones for this decade. The fuel barely exists yet. The entire licensed output of the United States to date is about 1,900 kilograms from one demonstration plant in Ohio, and a single 300-megawatt sodium core needs tonnes. And the regulator has to learn the physics before it can approve it. TerraPower's sodium reactor got its permit after years of pre-application meetings and 27,000 NRC staff hours. A pressurised water reactor on ordinary fuel has neither problem. The fuel is a commodity, and the physics is what the NRC has reviewed for seventy years. So for a data center that wants power this decade, the reactor is a small pressurised water reactor. The question is why nobody has one that works.

Two companies tried, and their failures are the map to the answer. NuScale spent seventeen years and $1.6 billion to certify a small light-water module. Its first customer, a group of Utah utilities, cancelled in 2023 when the power reached $89 a megawatt-hour, which loses to gas. Babcock & Wilcox spent about $400 million between 2009 and 2017 on mPower, a 195-megawatt module on the same principle, and shelved it when no utility would place an order. One failed on cost and one failed for lack of a buyer.

The cost failure has a physical cause. In a pressurised water reactor, the hot water from the core never touches the turbine. It runs through a steam generator, a heat exchanger that boils a separate loop of water into steam. That component has been the industry's biggest headache for fifty years, for two reasons. The first is size. A steam generator holds 3,000 to 16,000 tubes, stands up to 70 feet tall and weighs up to 800 tons, and each of the four at Vogtle weighs 1.4 million pounds. The second is the tubes themselves. They corrode, crack and wear, and because they are a barrier between the radioactive water and the outside, a leak is a safety event. Tube problems have caused about a quarter of the fleet's unplanned outage time, replacements have cost hundreds of millions of dollars per plant, and they helped shut down Trojan and San Onofre early. For a small reactor the size problem is the one that decides the economics. A small module puts the steam generator inside the reactor vessel, which is safer because there is no large pipe to break, but it means the steam generator now sets the size of the vessel. NuScale's is a coil of tubes wrapped around the core, and the vessel that holds it is 76 feet tall and weighs about 700 tons for 77 megawatts. The vessel sets the size of the containment. The containment sets the size of the building and the concrete. So a small reactor built the old way is still a large object, and it gives up the economies of scale of a big plant without gaining the economies of a factory. Shrink the steam generator and everything downstream of it shrinks with it. That is the component this stop leads to, and two companies on the map have taken it on from opposite ends.

Apollo Atomics shrinks the steam generator. The idea comes from fifteen years of work at MIT under Koroush Shirvan, who designed a compact integral reactor in 2010 and has since tested printed-circuit steam generators with the Naval Nuclear Laboratory, and who advises the company. A printed-circuit heat exchanger replaces the bundle of tubes with a stack of metal plates. Channels a few millimetres wide are etched into each plate, the plates are fused into one solid block, and the reactor water runs through one set of channels while the steam water runs through the other. Because the channels are small, the block holds far more heat-transfer surface in the same volume than tubes can. Because it is fused rather than welded, it has no tubes to crack and no joints to fail, which takes on the second headache as well. Where a conventional generator is hand-built and several storeys tall, Apollo's is about the size of a person and can be mass-produced. The company says it packs about ten times the power density and cuts the generator to about five percent of the size. Follow that through the chain above and the vessel, the containment and the building shrink with it. Apollo claims a forty-fold smaller footprint, small enough to assemble reactor and steam generator together in a factory, carry on a truck and install in under two years, in sizes from 10 to 300 megawatts. Everything else is left alone. Standard fuel, light-water physics, so the fuel is available and the regulator is asked to evaluate one new component. The company has run a 40-kilowatt demonstration reactor with MIT, filed its regulatory engagement plan this year, is seeking NRC authorisation of its fuel configuration by the end of 2026, reports more than 20 gigawatts of signed letters of intent, and raised $31 million in August, $26 million of it equity, to build a one-megawatt demonstrator ahead of a planned commercial plant in 2028. Its stated target is power at three cents a kilowatt-hour, which is a bet on beating gas rather than matching it. The gap is scale. That demonstrator is one three-hundredth of the largest planned product, and the forty-fold figure is the company's own.

Applied Atomics fixes the other failure. mPower did not fail on physics or on cost estimates. It failed because in 2017 no utility would buy a small reactor. That buyer has changed. A data center will sign a twenty-year contract for firm power and pay a premium for it, and it does not want to own the plant. In June Applied secured an exclusive land-based licence to the mPower design, with BWXT keeping the IP and the right to manufacture every component. Applied builds, owns and operates the plants, from 100 megawatts to a gigawatt, on the customer's land, and sells the power. The customer never touches a reactor. The release is careful to say Applied will "resume" design certification, and that word carries weight. B&W;'s pre-application work with the NRC ran from 2009 to 2014 and it never filed the certification application itself, so the licence buys eight years of engineering and the preserved test facilities, and the regulatory work restarts from the pre-application stage where it stopped. The licence is still the ownership move. It is a document no competitor can route through, and the build-and-operate model is what converts the document into revenue.

These are two different plays on the same reactor, and both get faster if the regulator's clock keeps moving. It has started to, with an eighteen-month deadline for new reactor decisions, a new licensing framework in effect since April, and the mandatory hearing taken off the critical path in May.

Stop three: what stands between the fence and the chip?

Power arrives at the developer's fence at tens of thousands of volts. A chip runs on about one. Getting from one to the other takes a series of machines, and they fall into two groups.

The first group sits at the fence. Transformers bring the voltage down. Circuit breakers and switchgear protect the lines. The developer does not build these. It orders them from a global industry, and today that industry is years behind. A substation transformer takes 128 weeks to arrive. A high-voltage breaker takes 125. Switchgear takes 44. The second group sits inside the building. A transformer brings the voltage down again to 480 volts. A UPS turns that AC into DC to charge its batteries, then back into AC to feed the floor. Power supplies in the racks turn it into DC once more, and converters on the boards take it down to the chip. Every one of those conversions gives up a point or two of energy, and NVIDIA's own materials put the whole chain below 90 percent efficient. So the first group costs the developer years, and the second group costs a tenth of the power. Two companies on the map take one group each.

Start with the years. The machines at the fence are not hard to make. A transformer is copper wound around a steel core, and building one takes a few months. The years are queue, and the queue has a simple cause. American electricity demand was essentially flat for nearly two decades before 2020, growing less than a third of a percent a year. For two decades the industry that makes grid equipment sold only replacements, so it sized its factories for replacements and closed the rest. Then demand came back all at once, and the factories that were left could not keep up. The one American mill that makes the special steel for transformer cores cannot keep up either, so even a new American factory waits on imported steel. Imports now supply about 80 percent of American power transformers. The whole queue is one industry's memory of twenty flat years, cast in factory capacity.

Not every country had twenty flat years. Indian manufacturers kept building capacity through the same period because their own grid kept growing, and by 2024 they had more of it than their home market needed. Anirudh Reddy, an IIT Madras engineer who had worked at Eaton and at the Indian EV maker Ather, saw the gap and founded Ayr Energy to close it. Ayr designs the equipment to American standards, orders the slowest components like bushings and tap changers ahead of demand, and has the Indian factories build to its specification, the way a contract manufacturer builds for a car company. It covers the whole first group, transformers, high-voltage breakers and medium-voltage switchgear. Medium-voltage units arrive in twelve to sixteen weeks and high-voltage units in forty to seventy-two, against three to five years from the incumbents. The order book passed $500 million in April and covers more than twenty gigawatts.

Run the arithmetic on that order book. Five hundred million dollars across twenty gigawatts is about twenty-five dollars a kilowatt. That sits inside the normal range for the equipment, from six to nine dollars a kilowatt for the largest transmission transformers to thirty-five to fifty for smaller substation units in North America, and the order book includes breakers and switchgear as well. So Ayr hands its customers two years and charges them ordinary prices. The time it saves is captured downstream, by the developer whose project starts earning sooner. That is the general condition of shortening. It also means the toll is temporary, because the incumbents are catching up. GE Vernova's electrification segment booked $2.4 billion of data center equipment orders in the first quarter of 2026, more than in all of the prior year. So Ayr's durable prize is elsewhere. Utilities buy only from vendors on their approved lists, getting onto a list takes years of testing and field acceptance, and a competitor cannot hire its way onto one. Shorten to enter, own to stay.

Now the tenth of the power. The conversions inside the building exist for one reason. The grid delivers AC, and it delivers AC because it was built to move power long distances to spinning motors. A campus that generates its own power on its own land needs neither the distance nor the motors. Solar panels make DC. Batteries store DC. Chips consume DC. Every conversion in between is there because the grid is in the middle, and if the grid is not in the middle they can go.

Voxel Energy removes them. Its founders came out of Tesla. Casey Spencer worked on Autopilot hardware and the Model 3 lines. Max Pfeiffer spent seven years building an EV company on second-life batteries and holds the patent on Voxel's architecture. Evan Schmidt brings a decade of data center construction. They build data centers with their own solar and battery generation, run as one DC system from the panel to the chip. Solar feeds a 380-volt DC bus. The batteries connect to it through DC-DC converters. The rack power supplies draw from the bus and step it down to the chip, which is the stage stop four is about and the one no architecture escapes. Between the panel and the rack, Voxel says that leaves two conversions at about two percent loss each. The same solar-and-battery plant built from grid parts runs through six or more, and the chained losses can reach a quarter of the power. The batteries are second-life Tesla packs bought from dismantlers, and they are sized so large against the load that the surges of a training run, which would destabilise an AC grid, barely register. Deleting a conversion removes its loss and its equipment queue at the same time, so Voxel runs all three forms of the shorten move at once. Removed losses, removed steps, and removed waiting, because with no grid connection there is no interconnection queue. A prototype is running off-grid compute, thousands of acres are under contract, and the first campus in New Mexico targets hundreds of megawatts through 2028. What it compounds into is what Pfeiffer brought with him. Years of degradation data on used packs and the dismantler relationships that supply them decide whether a second-life battery is an asset or a liability, and a later entrant would have to spend those years to find out.

Stop four: what must exist for NVIDIA's 800 VDC roadmap to happen?

Earlier in this essay I said the industry announces its own future in writing, years ahead, and then finances the companies that must exist to meet it. This stop is one of those companies, and the calendar it arrived on is NVIDIA's 800 VDC architecture, published in May 2025 for megawatt racks due in 2027. In the full version of the architecture, grid power is converted once, from 13.8-kilovolt AC to 800 volts DC at the edge of the building, and 800 volts DC runs all the way to the rack's input. The step down from there to the roughly one volt a GPU runs on happens inside the rack. NVIDIA puts the gain at up to five points of end-to-end efficiency and up to 45 percent less copper. It also names the parts the transition needs, most of which its partners are still developing, the rectifiers and solid-state transformers at the power room, the busbars and connectors that carry 800 volts down the row, and the converters inside the rack that take 800 volts down to the chip. Google and Microsoft have since joined the work through the Open Compute Project, and more than eighty manufacturers are building to the specification.

As racks head toward a megawatt, the industry is raising the voltage delivered to the rack from 54 volts to 800. At 54 volts a megawatt draws close to twenty thousand amps, and the copper, heat and loss scale with it, so raising the voltage fixes the wiring. It also creates a new job inside the rack. Something has to step 800 volts down to the roughly one volt a GPU runs on, in as few stages as possible, because each conversion stage costs a few percent of the energy and a slice of rack space that could have held compute.

The ideal part for that job has to block around 1,200 volts and switch fast. Fast switching is what shrinks the magnetics and lets a design skip a stage. Silicon is too slow at that voltage. Silicon carbide holds the voltage comfortably and is the incumbent, but its switching losses keep designs in the low hundreds of kilohertz where gallium nitride runs near a megahertz. That gap is why onsemi's GaN vice president, Antoine Jalabert, told DCD that looking at NVIDIA's roadmap "there's no other way than to use GaN" at those power densities.

GaN is still undertapped here. It was expected to ramp in servers ten to fifteen years ago and instead it is in laptop chargers. Nearly every GaN transistor in volume production today is lateral, designed for a laptop brick. A lateral device holds its voltage across the surface of the chip, so every extra volt of rating costs die area directly, and the economics run out around 650 volts. The current also runs in a thin channel just under the surface, which is where heat concentrates and where reliability suffers. And a lateral device has no p-n junction, so it cannot avalanche and has no way to protect itself through a transient overvoltage, which matters on an 800-volt bus. Today's 800-volt designs therefore fall back to silicon carbide or stack several lower-voltage GaN parts in series.

Vertical GaN answers those problems at once. Run the current down through the crystal instead of across its surface. Voltage is then held by how thick the GaN layer is and current by its area, so a 1,200-volt rating stops costing die area, heat spreads through the bulk instead of sitting at the surface, and the structure can be built to survive overvoltage. That is the switch the 800-volt rack wants, first in the power supplies and bus converters that feed it.

It was demonstrated in labs fifteen years ago and is only now reaching customers. The reason it took this long is the wafer. Lateral GaN wins today because it grows a thin layer on a cheap 200-millimetre silicon wafer anyone can buy. A vertical device needs a thick, clean GaN layer, and for most of that time the only wafer it would grow on without cracking was bulk GaN, small, expensive and from a handful of suppliers. Two routes are now opening up, bulk GaN getting cheaper and larger, and engineered wafers that let thick GaN grow on standard 200-millimetre substrates. Which one wins is the thing to watch. Jalabert's own view is that the hard part is the epitaxy, growing the GaN on the wafer, and that American capacity for it is thin.

That first step-down inside the rack is where vertical GaN sits. It is the switch that has to hold off the 800-volt bus and switch fast enough to keep the converter small. The internal rack bus is still around 50 volts today and the direction of travel is 12 volts and then 6, which only tightens what that first stage has to do.

Vertical Semiconductor is the company building that switch. It spun out of MIT's Palacios Group, one of the world's leading gallium-nitride labs. The device comes from Josh Perozek's doctoral research in that lab, Tomás Palacios, who runs the lab, is a co-founder, and Cynthia Liao is the chief executive. Two American companies tried this before and both failed within the past three years, and the reasons matter. NexGen Power Systems closed its Syracuse fab in December 2023 and went bankrupt, and onsemi bought the fab and its IP a year later. Odyssey Semiconductor sold its assets to Power Integrations and dissolved in 2024. Both built on bulk gallium-nitride wafers, the first of the two routes above, and the incumbents now own what they built. Vertical took the second route. The company has demonstrated its devices on 8-inch wafers using standard silicon CMOS processing, rated from 100 volts to 1.2 kilovolts, and it goes to market fabless. That puts vertical GaN on the same manufacturing base as the rest of the industry instead of on a scarce specialty wafer, which is the difference between a device that works in a lab and a device a supply chain can make at volume. The company claims up to 30 percent efficiency gains in half the power footprint. It began sampling packaged prototypes at the end of 2025, with an integrated solution due this year. It raised an $11 million seed led by Playground Global, with Shin-Etsu as a strategic investor. Shin-Etsu is the Japanese company that makes the wafers a vertical device would be grown on, so the investor list answers the epitaxy question above. The roadmap creates the demand for this switch. It does not decide who supplies it. NVIDIA sets the date the 800-volt rack ships, and the component maker's job is to have a qualified part ready when it does.

Stop five: why doesn't the building have an operating system?

When power, cooling and compute are engineered this tightly, the building becomes a machine. Here is what the machine does. A training job runs thousands of GPUs in lockstep. Each step has a compute phase near full power and a communication phase near idle. Because the GPUs move together, the building's draw is a square wave, repeating once a second to once every ten. For 100,000 H100s the swing is about 67 megawatts at the grid connection, for the length of the job, and Microsoft, OpenAI and NVIDIA have measured it in production. Every one of those megawatts becomes heat within seconds, and the coolant has to keep up or the chips throttle.

Operators cannot predict that wave, so they provision for its peak and hold the rest back. The margin runs twenty to thirty percent of provisioned power. On a hundred-megawatt hall that is twenty megawatts that were permitted, bought, wired and cooled, then frozen.

Nobody can release that margin because nobody can see it. A data center runs on five systems, a building-management system for cooling, a power-monitoring system for switchgear, SCADA for industrial control, a network operations center for the IT gear, and DCIM for capacity planning. Five vendors, five protocols, no shared model of the building. The BMS controls but does not model. DCIM models but cannot control. They poll about once a second, the period of the swing they would need to see. A controller cannot release margin it cannot measure at the speed the load moves.

So the building needs software that does three things at once, and none of the five systems does all three. See every device across the five protocols at sub-second speed. Predict what the next step does to the electrical and thermal state. Act on setpoints, power paths and eventually GPU power caps so the margin becomes compute. Acting is the hard part, because writing to a breaker is where the liability lives.

Two things make this buildable now. The load became controllable, because NVIDIA's GB300 power shelf ships with programmable ramp rates, a power floor and capacitor storage that cuts the peak the grid sees by up to thirty percent. And the workload became predictable, because a training run's period is known to the scheduler before it starts.

Aravolta is building that software. It replaces the five systems with one, and the centre of it is a device it calls the Node, a virtual PLC that speaks to the installed hardware the way the vendors' boxes do. The Node finds every device on the network, matches it against more than 25,000 templates, and is monitoring in days rather than months. Control is off by default and switched on per device, so an operator can change a setpoint, switch a power path, or start a generator from a browser. The predictive layer is claimed and cannot be checked from outside. The product is aimed at brownfield halls being pushed to AI densities, where the margin is largest and the incumbent's system is already failing.

The shortening is recovered loss, megawatts bought and frozen as safety margin, handed back by software that makes the building predictable. The ownership is what an installed controller becomes, the system everything else is wired through. Greenfield belongs to Schneider, Siemens and Vertiv, so brownfield is the only wedge, and NVIDIA putting smoothing into the rack shelf shrinks the swing and part of this product with it. The bet is that a building still needs its own control layer even when the racks smooth their own power. Windows like this do not stay open.

Stop six: where does the heat go in a state that is running out of water?

Every joule that enters the building leaves as heat, so a one-megawatt hall is a one-megawatt heater. The first war over that heat, getting it off the chip, just ended. Direct-to-chip liquid cooling won, and Ecolab's $4.75 billion for CoolIT was the settlement. A closed window points downstream, and downstream of the chip is the site boundary, where the heat has to leave the property. The tools for that are mechanical chillers, which consume a fifth to a third of a facility's power, or evaporative towers, which use millions of gallons of water a year. The buildout concentrated in exactly the places that can least afford either. The Houston Advanced Research Center projects that Texas data centers will use about 25 billion gallons of water in 2025 and somewhere between 29 and 161 billion gallons a year by 2030, up to 2.7 percent of the state's total. The width of that range is itself a message about how little anyone knows, and water politics already shape permits.

So the open problem is precise. How do you reject heat in a hot, dry place without a compressor and with far less water than a tower?

Start with how the standard system works. A chiller is paired with an evaporative cooling tower. The tower sprays warm water over a large surface while fans pull air through it, some of the water evaporates, and evaporation carries heat away. That is where the limit sits. Evaporating water can only cool something down to the air's wet-bulb temperature. That is the temperature a wet surface settles at when air blows over it, the same reason a wet shirt feels cold in a breeze, and it depends on how much moisture the air already carries. In dry air the wet surface gets far colder than the air. In humid air it stays close to air temperature. On a hot Texas afternoon the wet-bulb temperature is still not cold enough for the chips, so a chiller finishes the job, and the chiller is the compressor that eats a fifth to a third of the facility's power.

Dew-point cooling gets below the wet-bulb limit with a trick of sequence. Incoming air is first sent down a dry channel, where it is cooled through a thin wall by a wet channel on the other side, without touching any water itself. Cooling air without adding moisture lowers its wet-bulb temperature, because cooler air can hold less water. A portion of that pre-cooled air is then turned into the wet channel, where it evaporates water and, having started colder, reaches a lower temperature than the original air ever could. That cold wet channel is what cools the dry channel. The rest of the air leaves cold and dry as the product, and the theoretical floor is the dew point, which in dry climates sits well below wet-bulb. The cost is that part of the air is spent as working air, and some water still evaporates, but there is no compressor anywhere in the loop and less water is used per unit of cooling than a tower needs. The drier the climate, the bigger the gap between the wet-bulb and dew-point limits, so the technique performs best precisely where the buildout went and where towers are politically hardest to permit. The boundary is real too. This is dry-air physics, built for West Texas and the desert Southwest. It is not for Houston, where summer humidity puts the dew point itself too high to be useful.

Madrone packages that physics for data centers. Its founders are Akshay Trikha, who built machine-learning systems at QuantumScape, and Erik Meike, who spent four years at Apple on power and thermal systems for the iPhone, and they design everything in-house, the thermodynamics, the heat exchangers, the electronics and the control software. The unit sits outside the building in something like a shipping container and outputs cold water, so it is indifferent to what chips or racks are inside. In Texas it claims to cool with 30 percent less power and water than a chiller and tower. The company scaled from a one-kilowatt prototype to a hundred kilowatts during its YC batch this spring, has a one-megawatt unit next, which is the size hyperscalers say they need before procurement conversations start, and its first investor is a Texas developer building $20 billion of capacity, which gives it a site for the first pilot rather than capital alone.

Its move is deleted steps and recovered loss at once. The compressor is gone from the heat's exit path. A share of the facility's power is handed back to compute. And in a water-constrained county the saved water converts into the scarcest input of all, permit approval.

Stop seven: three doors through the memory wall. Which opens first?

The walk does not end at the chip's edge. It continues inside the package, and the same logic applies. Start with what a GPU does to produce one token. It reads every weight the model uses for that token out of memory, does a small amount of arithmetic on each one, and writes the result back. Frontier models today are mostly mixtures of experts, and the open ones publish the numbers. DeepSeek-V3 has 671 billion parameters and uses 37 billion of them on any given token. Stored at one byte each, those 37 billion weights are 37 gigabytes. NVIDIA's current top chip, the Blackwell Ultra, can pull 8 terabytes a second out of its memory and can do 5 petaflops of arithmetic at that precision. Reading 37 gigabytes at that speed takes about four and a half milliseconds. The arithmetic on those bytes takes about fifteen microseconds. For a single stream of tokens the chip works for fifteen microseconds and waits for four and a half milliseconds. Serving many users at once spreads the reading across more tokens, but every user's context has to sit in memory too, and that context now runs to hundreds of thousands of tokens. Either way the chip is waiting on bytes, and while it waits it draws power and produces nothing.

That imbalance is old and it is widening. Over twenty years peak server compute grew about 60,000-fold while memory bandwidth grew 100-fold. Compute scaled with transistor density. Moving bytes scaled with wires, pins and packaging, which do not shrink the same way. The mixture-of-experts design makes it worse on purpose. It saves arithmetic, because only 37 of the 671 billion weights fire per token. It saves no memory, because all 671 billion have to stay resident in case the next token calls them. So the job at this stop is bytes. More bytes per second into the compute die, and more bytes held within its reach.

The part that does that job today is high-bandwidth memory, and it is worth seeing what it physically is, because its three limits are the three places the companies at this stop attack. An ordinary memory channel talks to a processor over 64 wires. An HBM stack piles twelve memory dies on a base die and drills thousands of copper vias down through the silicon to connect the floors. It talks to the processor over 1,024 wires. A thousand wires cannot be routed across a circuit board. So the stack and the GPU sit millimetres apart on a slab of silicon called the interposer that carries the wiring between them. Blackwell Ultra carries eight such stacks ringed around its two compute dies, for 288 gigabytes. Three limits follow from that construction. The first is bandwidth. It is wires times speed per wire, and the number of wires is fixed by how much edge the compute die has to ring stacks around. The second is capacity on the package. It is stacks times dies per stack. Every die added to a stack multiplies the chance that one bad via kills the tower, which is a large part of why HBM has sold for four to five times the price of the same bits as server DRAM. The third is reach. Memory beyond the package can only be borrowed from another GPU over a copper link, which reaches across a board or a rack and no further, and every hop costs power and time. So a model that does not fit on one chip is split across several, and each of them mostly waits. Three companies make these stacks. The packaging step that marries them to the GPU was reported fully booked at the end of 2025. It is the cleanest physics-made chokepoint on the flow, and its seat is taken.

The software route around the wall, reusing the context cache across requests so that the same memory serves more tokens, was priced as solved when RadixArk seeded at $400 million. The other interesting approaches are the physical routes, and each of the three limits above has one. Change the wire that reaches beyond the package. Change the package itself. Or change the height of the stack. One young company on the map holds each, and three things make this the moment they are arriving. The CXL standard added memory pooling in its second version in 2020 and memory sharing in its third in 2022, which gave a processor a standard way to reach memory that is not its own. NVIDIA put co-packaged optics on its own switch roadmap in March 2025, which told every supplier that light inside the rack is coming. And the next memory generation, HBM4, allows sixteen-high stacks where the first products are twelve-high, which pushes the cost of stacking further up the same curve.

The first door changes the wire, and it answers the reach limit. Today CXL runs over copper, and at its data rates a copper link reaches across a board or a short cable and no further. Piris Labs carries CXL over light instead. An optical link holds its signal for about ten metres, which spans a rack and its neighbours, so the GPU servers across them can reach one shared pool of memory. A model that does not fit on one chip no longer has to be split across several to borrow theirs. The company has claimed five times lower latency and half the cost per token against the copper path. What makes the company interesting to me is who is making the claim. Ali Khalatpour, the chief executive, earned his MIT doctorate on terahertz lasers. He then built the laser at the heart of NASA's GUSTO balloon telescope. His co-founder Keyvan Moghadam led launch teams at Meta and X. The company is in YC W26 and holds a federal SBIR contract. When Lumilens raised its round in August, Khalatpour wrote publicly that moving more data onto optics solves nothing by itself, and that the hard part is the power it takes to generate and modulate the light. That is the cost that decides whether optics reaches inside the rack at all, and a founder who leads with it is building for the buyer's real constraint. What Piris is hostage to is adoption. CXL has to become the way GPUs reach memory, and the optics giants that Lumilens's round says are now funded have to leave that corner alone.

The second door changes the package, and it answers the bandwidth limit. If the thousand copper wires on the interposer are what fix bandwidth, replace them with light inside the package, where a waveguide carries far more bits per millimetre of edge than a copper trace. Volantis does this with a different light source from the rest of the industry. The company says that instead of silicon photonics it uses directly modulated lasers coupled into densely parallel optical waveguides, and it integrates compute dies, memory and the optics at wafer scale. What used to be a rack of servers wired together becomes one package. The company claims more than ten times the memory capacity and bandwidth the electrical package allows. I take the claim seriously because of Roy Meade, the chief technology officer. He led HBM development at Micron and was the first employee at Ayar Labs, so he has built both halves of the wall he is now trying to remove. Volantis's early backers include Alexandr Wang and Trevor Blackwell, and Sam Altman backed it as far back as 2022. Photonic computing has broken hearts before, and the reason is worth a moment. The earlier wave of optical chip companies tried to do the arithmetic itself with light, multiplying numbers by passing beams through modulators. Light is poor at that. It is analogue, so precision is limited, and every answer has to be converted back to electrical form, which costs the energy the optics were meant to save. Meanwhile electronic arithmetic kept getting cheaper every year, so the target moved. Volantis asks light to do only the thing it is good at, which is carrying bytes. A waveguide moves far more bits per second through a millimetre of edge than a copper trace, loses almost nothing over distance, and does not heat up as the data rate rises the way copper does. The arithmetic stays electronic. For twenty years arithmetic was the scarce thing, and a photonic chip had to win at arithmetic to matter. This cycle the scarce thing is bytes, and a photonic package only has to win at moving them. What Volantis is hostage to is yield. A package built from many lasers is only as good as the worst one.

The third door changes the height, and it answers the capacity limit. Stacking is how HBM gets its capacity, and the cost of stacking is what caps it, because every layer is a full lithography pass plus a bonding step. Atum Works builds a machine that prints three-dimensional metal structures directly, at 100-nanometre resolution across a whole wafer, the kind of structure that includes the vertical wiring between stacked chips. Its argument is that a 3D process scales in complexity with the materials used rather than with the number of layers, so stacking stops getting more expensive with height. If that holds, the four-to-five-times premium on stacked memory is a manufacturing accident rather than a law, and a sixteen-high stack stops carrying a penalty. The founders, Lucas Pabarcius, Matteo Kimura and Malcolm Tisdale, came out of Caltech and NASA. The advisory board includes TSMC's chief scientist, the director of Caltech's nanoscience institute and an Intel Fellow, which for a tool company is the right board, because those are the people who decide whether a new tool enters a fab. Atum went through YC in spring 2025, has a working prototype in Mountain View, and holds a letter of intent with NVIDIA for co-development. That is not production. It is how packaging relationships with NVIDIA start. What Atum is hostage to is the qualification cycle. A fab adopts a new tool over years, and a startup has to be funded across all of them.

Which door opens first is not knowable yet. The three are not exclusive, since a pooled optical memory fabric, a photonic package and a cheaper way to stack could all ship into the same rack. What they share is the move. A GPU stalled on memory burns full power producing nothing. Each of these widens the path to memory, so each recovers that loss on every accelerator it touches.

Stop eight: what does a GPU-hour actually buy?

The last stop is the other heavy end of the dumbbell, where silicon meets software. Start with a GPU that is racked, powered and not earning, because that happens far more than the price of the chip suggests. A study of a large GPU cluster published in April measured how often a GPU had a program loaded and was drawing elevated power while nothing ran. The answer was about a fifth of all execution time. For online serving it was 61 percent. When the authors replayed public industry serving traces the share ran from 14 to 76 percent. The reason is the shape of the work. A serving GPU receives a request, works for a fraction of a second and waits for the next one, and the gaps between requests run several seconds. The chip draws power the whole way through. A GPU-hour costs the same whether or not tokens come out of it, so the only lever at this stop is how many tokens an hour produces. That number is two numbers multiplied. The share of the hour the chip is actually working, and the rate it produces while it works. The companies here split the same way. The first group raises the share. The second raises the rate. And one company prices the hour, which is a separate story and comes last.

Start with the share, because that is what the study measured. The obvious fix for a chip that waits is to give it to another job while it waits and take it back when the next request arrives. Every computer since the 1960s has done this. An operating system holds several programs in memory at once, switches between them when one is waiting, and moves them around as it needs to. In production a GPU today mostly runs the way a mainframe ran before that. One program owns the whole machine, and when that program is waiting the machine waits with it. Two things stop the fleet from fixing this. The first is memory. The model that is waiting still occupies the chip's memory with its weights and its context cache, so there is no room to hold a second model that could use the gap. The second is that the job cannot leave. A running GPU job's state is spread across up to 288 gigabytes of memory per chip, the kernels in flight, and the communication handles that bind an eight-GPU job together, and none of that can be frozen and moved. So the chip cannot take on a second job, and the first job cannot be moved to a chip that is busy. The idle hours spoil where they sit. The two missing pieces are the two oldest ideas in operating systems, holding more than one program at a time and moving a program while it runs, and one company on the map is building each.

OpenInfer builds the first. Its control plane, Weave, oversubscribes the chip's memory, in the company's own description. It keeps more models resident on each device than would fit if every one were fully loaded, and it swaps between them as requests arrive. That is what an operating system does with a computer's memory, and it means one model's gap can be filled by another model's request. Weave also treats a session rather than a request as the unit of work, because the work is changing shape. Every serving platform was built for chat, a short request that finishes in seconds. The unit of demand is becoming a session that runs for minutes or hours, holds a growing context cache, and swings between bursts of GPU work and long stretches of CPU-side coordination between tool calls. To the GPU those stretches look exactly like the gaps in the study, so a serving layer that understands the session can plan around them. Its chief business officer, Kam Eshghi, co-founded Lightbits Labs, which pioneered NVMe over TCP storage. That history is relevant. Lightbits turned a storage device tied to one machine into a pool the whole fleet could draw on, which is what OpenInfer is trying to do to GPUs. Cedana builds the second. It interposes at the driver level to capture the full state of a running GPU job and restore it on another machine, so a job can be paused, moved and resumed while it runs. Virtual machines have been moved this way for twenty years, and that one capability is what let cloud providers pack servers full and sell the spare. GPUs have not had it, which is why a cluster today cannot be defragmented and spare capacity cannot be reclaimed without killing a job. Cedana came through YC in 2023. Both companies shorten, and what each could own is the same material, dependency. Once a fleet schedules through a control plane or checkpoints through a migration layer, ripping it out means downtime, and infrastructure software that reaches that point tends to stay.

There is a third way to raise the share, and it is to shrink the gap itself. In an agentic session much of the gap is time spent on the CPU, parsing a tool's output, deciding what to call next, and preparing the next request. A faster CPU for that work shortens every gap the GPU sits through. Nuvacore is designing that CPU from scratch. Its founders, Gerard Williams III, John Bruno and Ram Srinivasan, designed the cores and systems behind Apple's M-series chips. They then founded Nuvia, which Qualcomm bought for $1.4 billion. Sequoia backed the new company, which came out of stealth in April. There is no silicon yet, and the natural customer is a hyperscaler, so this is the longest and most expensive bet at the stop. It is also the only one whose position, if it earns it, would be made of CPU design IP rather than software dependency, which is the material this team already turned into a moat once.

Now the rate, because the hours the chip is working are not full either. When Meta trained Llama 3 on sixteen thousand H100s, with one of the most capable infrastructure teams in the world tuning the run, the GPUs reached 38 to 43 percent of their theoretical arithmetic. Some of the rest was lost to memory, which is stop seven's problem. Some was lost to waiting on other GPUs. And some was lost to software, which is this stop's problem, and that software is the kernels. A model is a graph of a few dozen kinds of operation, and each kind needs a kernel, a small program tuned to one chip's arithmetic units and memory hierarchy. How well a chip runs a model is how well its kernels fit. CUDA is nearly two decades of those kernels written for one company's chips, and even on NVIDIA hardware the stock kernels leave performance behind on any workload they were not tuned for. On a new chip there are no kernels at all. So a new accelerator pays a tax before its first production customer, months to years of scarce specialist work per architecture, and the hardware is inventory until the tax is paid. That tax is a large part of why chips that beat NVIDIA on paper have not beaten it in the market. What changed is that writing kernels is one of the few kinds of specialist work models are already provably good at, so the cost of paying it is falling fast. Three companies turn that into a business.

Wafer points it at hardware already racked, which is the same hardware the first half of this stop was about. Its agents learn a workload's traffic and search across the model, the engine, the kernels and more than a hundred serving settings. A change ships only when it returns identical outputs and measures faster. One of its results shows why this belongs next to stop seven. Kimi K3 is a 2.8-trillion-parameter mixture of experts whose weights alone exceed 1.5 terabytes. Wafer fit it on a single eight-GPU AMD MI355X node because that chip carries more HBM. The NVIDIA comparison, B200s with 192 gigabytes each, needed sixteen GPUs across two nodes, and under the rental prices Wafer assumed the AMD node came out ahead per dollar. NVIDIA's newer B300 carries 288 gigabytes and fits the model on eight GPUs as well, so the gap is a generation rather than a standing disadvantage. Memory capacity still decided the economic winner, and it took software to find that out. Wafer raised a $40 million Series A in early September co-led by Marathon and Chemistry with AMD Ventures in the round. Infinity points it at chips that have no kernels yet. Its agent, Ignition, writes the kernels for a new chip and tests them on the hardware. Then it rewrites them as performance data comes back. In its published work with d-Matrix the company says Ignition reached 92 percent of the Corsair chip's theoretical peak ten hours after first touching the hardware, and had three frontier models running end to end within ten days. Those are the company's own numbers, and if they hold, work that has taken chip companies years took ten days. It raised a $15 million seed at a $100 million valuation in July from Touring Capital with researchers from OpenAI and Anthropic among the angels. Kernelize builds the road so that the tax stops being owed at all. Triton is the open kernel language the industry has settled on, and PyTorch's own compiler emits it by default, which makes every PyTorch user a Triton user without their knowing it. Kernelize's founder, Simon Waters, led AMD's Triton compiler team. The company builds the extensions that map Triton onto a new architecture, so a new chip plugs into the language and inherits everything already written in it.

Whether any of the three compounds into a position depends on standardisation. A bring-up tool is a vendor until the industry routes through it by default, and a compiler layer the ecosystem standardises on is a chokepoint made of process. Triton is how that standard formed, which is why Waters having led it at AMD is the detail that matters. NVIDIA does not lose from this. Its lock is hardware and supply, the HBM contracted years ahead and the rack systems nobody else ships. What weakens is the software barrier that made everyone else's chips unusable, and as it falls every new accelerator becomes usable and every installed GPU gets cheaper to run.

The two groups above change how many tokens an hour produces. The last company at this stop changes how the hour is priced, and the reason it belongs here is who owns the hours. Many of the GPUs being installed today are bought by companies that rent them out, and many of those companies borrowed to buy them. Their cost is set by the price of power and the depreciation of the chip. Their revenue is the price of a GPU-hour. The spread between those two prices is their margin. And the sellers are converging on one shape. Andrawes Bahou, Compute Desk's founder, describes a market where everyone buys hardware, rents GPUs, and sells GPU-hours, often to their own competitors, with no differentiation left except how well you secure supply. That is the definition of a commodity, and a commodity without a benchmark is a market waiting for one. Power has had a public benchmark price for decades. Reserved compute has had none. Reserved compute trades in blocks worth hundreds of millions of dollars and a large deal still closes over email. On a ten-thousand-GPU contract running for a year, a difference of one cent per GPU-hour is close to $900,000, and there is no reference price to settle the argument and no neutral place for the money to wait. Compute Desk publishes daily GPU rental price indexes built from verified private transactions, carried on Bloomberg and Refinitiv. On September 3 it announced a partnership with Nodal Exchange to launch CFTC-regulated compute futures that settle against those indexes, due this year. Nodal is a power exchange that offers the world's largest set of locational power futures, and that choice is the interesting part. A company that rents out GPUs will be able to hedge the compute it sells and the power it buys on the same exchange. That lets it lock in the spread that is its business. A lender to that company can underwrite against a public benchmark. Bloomberg framed the launch as a rival to contracts CME and ICE are pursuing. Every commodity that matured got its reference price first, and whoever published it tended to keep the seat for decades. That is a position made of process, and it makes Compute Desk the one company at this stop whose move is owning rather than shortening.

The three groups are three answers to the same question. A GPU-hour is paid for in full. The first group makes more of the hour usable. The second makes the usable part produce more. And the index prices what the hour is worth once those two numbers start to move. The buyer for all of it is anyone who already owns GPUs, and that is why this end of the flow is where founders are crowding.

What does the Spacing Guild know about chokepoints?

The Spacing Guild never mined a gram of spice. It monopolised transport, the one stage of that universe's flow nobody could route around, and it converted the position into a toll on everything that moved. But the Guild's monopoly had a dependency of its own. Its navigators could only steer a ship by consuming spice, and spice came from one planet. That put the Guild's input in someone else's hands. Whoever controlled the planet controlled the spice, and whoever controlled the spice controlled the Guild, however complete its monopoly on transport looked. That is how the story ends. Paul Atreides takes the planet, threatens to destroy the spice, and the Guild has no choice left. The lesson is that a chokepoint is only as strong as whatever sits upstream of it. Chokepoints nest. Enriched fuel sits upstream of the reactor licence. HBM sits upstream of every accelerator. Energized land sits upstream of everything. And upstream of all of it sit two things this essay has left alone. One is permission to build, which is being renegotiated county by county, with opponents blocking or delaying $130 billion of projects in the first quarter of 2026 alone and New York pausing permits for large data centers in July. The other is the efficiency of the models themselves, since the price of a fixed level of performance has been falling between nine and nine hundred times a year, and every queue on the walk shortens if that outruns demand. So when a company on this flow says it owns a point nobody can route around, the thing to check is what its own position hangs from. That is the lens I ended the walk with, and it is why the walk started where it did.

So go back to the two contracts at the top. Anthropic paid $271 a megawatt-hour for a place at the front of a five-year line. Reflection paid the equivalent of nearly $5,000 for the line's absence. Both bought the same electrons, and the difference between them was time. The walk was an attempt to find where that time actually lives, and the answer turned out to be specific. It lives in a utility study that takes years to start. In a steam generator too large to build fast. In a transformer queue set by a factory sized for a flat decade. In a conversion stage that did not need to exist. In a safety margin nobody measured. In a chiller in a county with no water to spare. In a chip that waits four milliseconds for bytes it will use for fifteen microseconds. In a GPU that is racked, powered and idle for a fifth of its hours. Each of those is a place on the flow where a young company can stand, and at each of them the founders in this essay are doing one of two things. They are shortening the flow, by removing a loss, a step or a wait. Or they are owning a point on it that the flow cannot route around.

What I took from the map is that the second move is rarely available to a new company on its own, because the points made of physics and law are mostly claimed. So almost everyone enters by shortening. The companies I keep coming back to are the ones whose shortening leaves something behind, a dataset, a qualified-vendor status, an installed control plane, a process the industry routes through by default. That residue is what a position is made of.

Herbert's line was about a universe that ran on one substance from one planet. Ours runs on one flow, from a turbine to a token, and the same rule holds. When everything depends on a flow, power belongs to whoever stands where it narrows. The work is knowing where a company stands on it, what it can shorten from there, what it could come to own, and what stands upstream of it that it cannot see around.

The electrons must flow.

The Energy to Compute Flow Terminal, the 450+ company map this essay is built on, is public and updated continuously. If you're building at any narrowing of this flow, or if you think any claim here is wrong, I want to hear from you. The map improves when it's argued with.